

Homogeneous system

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= 0 \\a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= 0 \\&\dots \\a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= 0\end{aligned}$$

This homogeneous system of have at least 1 sol., that is $(0, 0, 0)$. ^{Trivial solution}

$\therefore$ A homo. system will always be consistent.

It either have a unique sol. OR infinitely many sol.

Eigenvector

$\sim$ Vec. whose "direc." does not change [stays on the same span] after a particular transformation. ^{Opposite direc. also is same span}

$$\boxed{A \underline{v} = \lambda \underline{v}} \implies$$

E. Vec
Trans. Matrix E. Val

$$A \underline{v} - (\lambda I) \underline{v} = 0$$

$$\boxed{(A - \lambda I) \underline{v} = 0}$$

Beuz obviously the zero-vec. is always an e-vec. w e-val = 0

* This is a homo-system. We do not want the unique solution [$\underline{v} = 0$], so we set this system to have infinitely many sol. by making $(A - \lambda I)$ not invertible

$$\therefore \det(A - \lambda I) = 0$$

Characteristic Eqn

* No. of e-val = Degree of Characteristic Eqn (might be repeated)

Algebraic Multiplicity

How many times λ appears

Geometric multiplicity

How many linearly independent e-vec correspond to a particular λ

Example:

$$(\lambda - 1)^3 (\lambda + 3) (\lambda - 7) = 0$$

No. of e-val : 5 [1, 1, 1, -3, 7]

No. of distinct e-val : 3 [1, -3, 7]

Algebraic multiplicity of $\lambda = 1$ is 3

$\lambda = -3$ is 1

$\lambda = 7$ is 1

Example 10

Find the eigenvalues and eigenvectors of the matrix $\mathbf{B} = \begin{pmatrix} -2 & 1 \\ 6 & 3 \end{pmatrix}$.

[**Note:** Use each eigenvalue to produce a corresponding eigenvector. Remember that any non-zero scalar multiple of an eigenvector is also an eigenvector corresponding to the same eigenvalue.]

$$\left(\text{Ans: } \lambda = -3, \begin{pmatrix} 1 \\ -1 \end{pmatrix}; \lambda = 4, \begin{pmatrix} 1 \\ 6 \end{pmatrix} \right)$$

Solution

$$\mathbf{B} - \lambda \mathbf{I} = \begin{pmatrix} -2-\lambda & 1 \\ 6 & 3-\lambda \end{pmatrix}$$

$$\det(\mathbf{B} - \lambda \mathbf{I}) = 0$$

$$\Rightarrow (-2-\lambda)(3-\lambda) - 6 = 0$$

$$\Rightarrow \lambda^2 - \lambda - 6 - 6 = 0$$

$$\Rightarrow \lambda^2 - \lambda - 12 = 0$$

$$\Rightarrow (\lambda - 4)(\lambda + 3) = 0$$

$$\Rightarrow \boxed{\lambda = 4} \text{ or } \boxed{\lambda = -3}$$

$\therefore$ The eigen values of $\mathbf{B}$
are 4 and -3

$$(\mathbf{B} - \lambda \mathbf{I}) \vec{x} = \vec{0}$$

$$\Rightarrow \begin{pmatrix} -2-\lambda & 1 \\ 6 & 3-\lambda \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \text{ where } \vec{x} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\Rightarrow \begin{cases} (-2-\lambda)x + y = 0 \\ 6x + (3-\lambda)y = 0 \end{cases}$$

For $\lambda = 4$

$$\begin{cases} -6x + y = 0 \\ 6x - y = 0 \end{cases}$$

There must be one free variable
for this system

Let $x = t$, then $y = 6t$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} t \\ 6t \end{pmatrix} = t \begin{pmatrix} 1 \\ 6 \end{pmatrix}$$

But becaz we want
a specific soln

For $\lambda = 4$, an eigenvector
is $\begin{pmatrix} 1 \\ 6 \end{pmatrix}$

For $\lambda = -3$

$$\begin{cases} x + y = 0 \\ 6x + 6y = 0 \end{cases}$$

$$x = -y$$

For $\lambda = -3$,

an eigenvector is $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$

Complex e.vec & e.val

$$A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

$$\det(A - \lambda I) = 0$$

$$\det \begin{pmatrix} \cos \theta - \lambda & -\sin \theta \\ \sin \theta & \cos \theta - \lambda \end{pmatrix} = 0$$

$$(\cos \theta - \lambda)^2 - \sin \theta (-\sin \theta) = 0$$

$$\lambda^2 - (2 \cos \theta) \lambda + (\cos^2 \theta + \sin^2 \theta) = 0$$

$$\lambda^2 - (2 \cos \theta) \lambda + 1 = 0$$

$$\lambda = \frac{-(-2 \cos \theta) \pm \sqrt{4 \cos^2 \theta - 4}}{2}$$

$$= \cos \theta \pm \sqrt{\cos^2 \theta - 1}$$

$$= \cos \theta \pm \sqrt{-\sin^2 \theta}$$

$$\lambda = \cos \theta \pm i \sin \theta$$

For $\lambda = \cos \theta + i \sin \theta$

$$\begin{pmatrix} \cos \theta - \cos \theta - i \sin \theta & -\sin \theta \\ \sin \theta & \cos \theta - \cos \theta - i \sin \theta \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} -i \sin \theta & -\sin \theta \\ \sin \theta & -i \sin \theta \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\left(\begin{array}{cc|c} -i \sin \theta & -\sin \theta & 0 \\ \sin \theta & -i \sin \theta & 0 \end{array} \right) \xrightarrow{R_2 \rightarrow iR_2 + R_1} \left(\begin{array}{cc|c} -i \sin \theta & -\sin \theta & 0 \\ 0 & 0 & 0 \end{array} \right)$$

$\therefore v_2$ is a free variable, $v_2 = t$

$$(-i \sin \theta) v_1 - (\sin \theta) v_2 = 0$$

$$i v_1 + t = 0$$

$$v_1 = -\frac{t}{i}$$

$$= i(t)$$

$$\therefore v = \begin{pmatrix} t \\ it \end{pmatrix} = t \begin{pmatrix} 1 \\ i \end{pmatrix}$$

$\therefore \begin{pmatrix} 1 \\ i \end{pmatrix}$ is an eigen vector for
 $\lambda = \cos \theta + i \sin \theta$

1) Find eigenval and corresponding eigenvector for $B = \begin{pmatrix} 3 & 2 & 4 \\ 1 & 2 & 0 \\ 1 & -2 & 1 \end{pmatrix}$

$$B - \lambda I = \begin{pmatrix} 3-\lambda & 2 & 4 \\ 1 & 2-\lambda & 0 \\ 1 & -2 & 1-\lambda \end{pmatrix}$$

$$\det(B - \lambda I) = 0$$

$$(3-\lambda) \begin{vmatrix} 2-\lambda & 0 \\ -2 & 1-\lambda \end{vmatrix} - 2 \begin{vmatrix} 1 & 0 \\ 1 & 1-\lambda \end{vmatrix} + 4 \begin{vmatrix} 1 & 2-\lambda \\ 1 & -2 \end{vmatrix} = 0$$

$\vdots$

$$\lambda^3 - 6\lambda^2 + 5\lambda + 12 = 0$$

$$(\lambda + 1)(\lambda - 3)(\lambda - 4) = 0$$

$$\therefore \lambda = -1 \text{ or } \lambda = 3 \text{ or } \lambda = 4$$

$$(B - \lambda I)\vec{x} = \vec{0}$$

$$\begin{pmatrix} 3-\lambda & 2 & 4 \\ 1 & 2-\lambda & 0 \\ 1 & -2 & 1-\lambda \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

When $\lambda = -1$

$$\left(\begin{array}{ccc|c} 4 & 2 & 4 & 0 \\ 1 & 3 & 0 & 0 \\ 1 & -2 & 2 & 0 \end{array} \right)$$

$$\xrightarrow{\substack{R_1 \rightarrow \frac{1}{2}R_1 \\ R_1 \leftrightarrow R_2}} \left(\begin{array}{ccc|c} 1 & 3 & 0 & 0 \\ 2 & 1 & 2 & 0 \\ 1 & -2 & 2 & 0 \end{array} \right)$$

$$\xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 3 & 0 & 0 \\ 0 & 5 & -2 & 0 \\ 1 & -2 & 2 & 0 \end{array} \right)$$

$$\xrightarrow{R_3 \rightarrow R_3 - R_1} \left(\begin{array}{ccc|c} 1 & 3 & 0 & 0 \\ 0 & 5 & -2 & 0 \\ 0 & -5 & 2 & 0 \end{array} \right)$$

$$\xrightarrow{R_3 \rightarrow R_3 + R_2} \left(\begin{array}{ccc|c} 1 & 3 & 0 & 0 \\ 0 & 5 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$$\text{Let } z = t,$$

$$x + 3y = 0$$

$$x = -3y \text{ --- (1)}$$

$$5y - 2t = 0$$

$$y = \frac{2}{5}t \text{ --- (2)}$$

$$\text{(2)} \rightarrow \text{(1)}$$

$$x = -\frac{6}{5}t$$

$$\therefore \vec{x} = t \begin{pmatrix} -6/5 \\ 2/5 \\ 1 \end{pmatrix} = \frac{1}{5}t \begin{pmatrix} -6 \\ 2 \\ 5 \end{pmatrix}$$

$$\therefore \text{For } \lambda = -1, \text{ an eigenvector is } \begin{pmatrix} -6 \\ 2 \\ 5 \end{pmatrix}$$

$$\text{For } \lambda = 3$$

$\vdots$

$$\text{Eigenvector is } \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$$

$$\text{For } \lambda = 4$$

$\vdots$

$$\text{Eigenvector is } \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$$

Alternative Method

1) Find eigenval and corresponding eigenvector for $B = \begin{pmatrix} 3 & 2 & 4 \\ 1 & 2 & 0 \\ 1 & -2 & 1 \end{pmatrix}$

$$B - \lambda I = \begin{pmatrix} 3-\lambda & 2 & 4 \\ 1 & 2-\lambda & 0 \\ 1 & -2 & 1-\lambda \end{pmatrix}$$

$$\det(B - \lambda I) = 0$$

$$(3-\lambda) \begin{vmatrix} 2-\lambda & 0 \\ -2 & 1-\lambda \end{vmatrix} - 2 \begin{vmatrix} 1 & 0 \\ 1 & 1-\lambda \end{vmatrix} + 4 \begin{vmatrix} 1 & 2-\lambda \\ 1 & -2 \end{vmatrix} = 0$$

$$\lambda^3 - 6\lambda^2 + 5\lambda + 12 = 0$$

$$(\lambda + 1)(\lambda - 3)(\lambda - 4) = 0$$

$$\therefore \lambda = -1 \text{ or } \lambda = 3 \text{ or } \lambda = 4$$

$$(B - \lambda I)\vec{x} = \vec{0}$$

$$\begin{pmatrix} 3-\lambda & 2 & 4 \\ 1 & 2-\lambda & 0 \\ 1 & -2 & 1-\lambda \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

For $\lambda = -1$

$$\begin{pmatrix} 4 & 2 & 4 \\ 1 & 3 & 0 \\ 1 & -2 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{bmatrix} \vec{R_1} \cdot \vec{x} \\ \vec{R_2} \cdot \vec{x} \\ \vec{R_3} \cdot \vec{x} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This means that $\vec{x}$ is perpendicular to $\vec{R_1}, \vec{R_2}, \vec{R_3}$ since the dot product = 0

So to find $\vec{x}$, we just have to cross product

2 non-parallel vec. out of the $\vec{R_1}, \vec{R_2}, \vec{R_3}$

$$\text{Eigenvec.} = \begin{pmatrix} 4 \\ 2 \\ 4 \end{pmatrix} \times \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 - 12 \\ -(0 - 4) \\ 12 - 2 \end{pmatrix} = 2 \begin{pmatrix} -6 \\ 2 \\ 5 \end{pmatrix}$$

$\therefore$ For $\lambda = -1$, an eigenvector is $\begin{pmatrix} -6 \\ 2 \\ 5 \end{pmatrix}$ #

No. of e.vec. & e.val

2×2 matrix $\bar{w}$ real entries

- 2 real, distinct e.val $\bar{w}$ e.vec
- 1 repeated e.val $\bar{w}$ e.vec
- Pair of complex conjugate, $(a + bi)$ with complex e.vec
- 1 repeated e.val where every vec. is an e.vec [Dilation]

Repeated e.val for 3x3 matrix

$$A = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{pmatrix}$$

$$\det(A - \lambda I) = 0$$

$\vdots$

$$-\lambda^3 + 9\lambda^2 - 24\lambda + 20 = 0$$

$$(\lambda - 2)(-\lambda^2 + 7\lambda - 10)$$

$$(\lambda - 2)[-(\lambda - 5)(\lambda - 2)] = 0$$

$$\lambda = 2, \lambda = 2, \lambda = 5$$

For $\lambda = 5$,

$\vdots$

$$\text{E.vec.} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

For $\lambda = 2$

$$\begin{pmatrix} 3-2 & 1 & 1 \\ 1 & 3-2 & 1 \\ 1 & 1 & 3-2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \end{array} \right) \xrightarrow[R_3 \rightarrow R_3 - R_1]{R_2 \rightarrow R_2 - R_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$$\therefore v_1 + v_2 + v_3 \Rightarrow v_1 = -s - t$$

$$v_2 = s$$

$$v_3 = t$$

$$\therefore \underline{v} = \begin{pmatrix} -s-t \\ s \\ t \end{pmatrix} = s \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

Linearly Independent

$\therefore$ 2 linearly independent vec.
 $\Rightarrow \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$

Geometric multiplicity is 2.

This means we have an entire plane of e.vec. [Infinitely many]

Diagonalisation

$$\text{Let } A = \begin{pmatrix} 1 & 1/2 \\ 1 & 3/2 \end{pmatrix}$$

$$\text{Assume } A \text{ has } \lambda_1 = 2, \text{e.vec 1} = \begin{pmatrix} 1/2 \\ 1 \end{pmatrix} \implies A \begin{pmatrix} 1/2 \\ 1 \end{pmatrix} = 2 \begin{pmatrix} 1/2 \\ 1 \end{pmatrix}$$

$$\lambda_2 = \frac{1}{2}, \text{e.vec 2} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \implies A \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$\text{Let } V \text{ be } \begin{pmatrix} \text{e.vec 1} & \text{e.vec 2} \\ 1/2 & -1 \\ 1 & 1 \end{pmatrix}$$

$$AV = \begin{pmatrix} 1 & 1/2 \\ 1 & 3/2 \end{pmatrix} \begin{pmatrix} 1/2 & -1 \\ 1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{2} + \frac{1}{2} & -1 + \frac{1}{2} \\ \frac{1}{2} + \frac{3}{2} & -1 + \frac{3}{2} \end{pmatrix}$$

$$= \begin{pmatrix} 2(1/2) & \frac{1}{2}(-1) \\ 2(1) & \frac{1}{2}(1) \end{pmatrix}$$

$$= \begin{pmatrix} \text{e.vec 1} & \text{e.vec 2} \\ \frac{1}{2} & -1 \\ 1 & 1 \end{pmatrix} \underbrace{\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}}_{\text{Diagonal Matrix, } D}$$

$$\therefore AV = VD$$

$$\boxed{A = VD V^{-1}}$$

* For this formula to be valid, V has to be invertible.

This is useful for finding high powers of matrices

$$A^n = (VD V^{-1})(VD V^{-1})(VD V^{-1})(VD V^{-1}) \dots (VD V^{-1})(VD V^{-1})$$

Becomes Identity Matrix I

$$\boxed{A^n = V D^n V^{-1}}$$

* Note that $D^n = \begin{pmatrix} a^n & 0 & 0 & \dots & 0 \\ 0 & b^n & 0 & \dots & 0 \\ 0 & 0 & c^n & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \end{pmatrix}$, so its very ez to find.

Diagonalisation1. **Definition (Diagonalisable matrix):**

A square matrix $\mathbf{A}$ is said to be **diagonalisable** if $\mathbf{A} = \mathbf{PDP}^{-1}$ for some **invertible** matrix $\mathbf{P}$ and some diagonal matrix $\mathbf{D}$.

2. **Fact:** All $n \times n$ matrices which have n **distinct eigenvalues** are **diagonalisable**.3. **Fact:**

Let $\mathbf{A}$ be an $n \times n$ matrix with n distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ and corresponding eigenvectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$.

Let

$$\mathbf{P} = (\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3 \quad \dots \quad \mathbf{e}_n),$$

i.e., the **columns** of $\mathbf{P}$ are the n **eigenvectors** of $\mathbf{A}$, and

$$\mathbf{P} = \begin{pmatrix} \boxed{} & \boxed{} & \boxed{} \\ \vec{e}_1 & \vec{e}_2 & \vec{e}_3 \end{pmatrix}$$

$$\mathbf{D} = \begin{pmatrix} \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & 0 & \dots & 0 \\ 0 & 0 & \lambda_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n \end{pmatrix},$$

$$\begin{pmatrix} \boxed{1} & \boxed{4} & \boxed{7} \\ \boxed{2} & \boxed{5} & \boxed{8} \\ \boxed{3} & \boxed{6} & \boxed{9} \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix} = \begin{pmatrix} 1\lambda_1 & 4\lambda_2 & 7\lambda_3 \\ 2\lambda_1 & 5\lambda_2 & 8\lambda_3 \\ 3\lambda_1 & 6\lambda_2 & 9\lambda_3 \end{pmatrix}$$

$\mathbf{e}_1\lambda_1 \quad \mathbf{e}_2\lambda_2 \quad \mathbf{e}_3\lambda_3$

i.e., $\mathbf{D}$ is a diagonal matrix whose **diagonal entries** are **eigenvalues** of $\mathbf{A}$ that correspond, respectively, to the eigenvectors in $\mathbf{P}$.

$$\mathbf{AP} = \begin{cases} \lambda_1 \mathbf{e}_1 \\ \lambda_2 \mathbf{e}_2 \\ \lambda_3 \mathbf{e}_3 \end{cases} \therefore \mathbf{AP} = \mathbf{PD}$$

Then it *must* be true that $\mathbf{P}$ is invertible, and that $\mathbf{AP} = \mathbf{PD}$. Hence, $\mathbf{A} = \mathbf{PDP}^{-1}$.

4. Note that

$$\mathbf{D}^m = \begin{pmatrix} \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & 0 & \dots & 0 \\ 0 & 0 & \lambda_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n \end{pmatrix}^m = \begin{pmatrix} \lambda_1^m & 0 & 0 & \dots & 0 \\ 0 & \lambda_2^m & 0 & \dots & 0 \\ 0 & 0 & \lambda_3^m & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n^m \end{pmatrix}$$

so it is easy to find powers of $\mathbf{D}$.

5. **Finding powers of square matrices:**

Consider a diagonalisable square matrix $\mathbf{A}$ that can be written in the form $\mathbf{A} = \mathbf{PDP}^{-1}$.

Suppose that we wish to determine $\mathbf{A}^{20}$.

Then

$$\begin{aligned} \mathbf{A}^{20} &= (\mathbf{PDP}^{-1})^{20} \\ &= (\mathbf{PDP}^{-1})(\mathbf{PDP}^{-1})(\mathbf{PDP}^{-1}) \dots (\mathbf{PDP}^{-1})(\mathbf{PDP}^{-1}) \\ &= \mathbf{PD}(\mathbf{P}^{-1}\mathbf{P})\mathbf{D}(\mathbf{P}^{-1}\mathbf{P})\mathbf{D}(\mathbf{P}^{-1}\mathbf{P}) \dots \mathbf{PD}(\mathbf{P}^{-1}\mathbf{P})\mathbf{DP}^{-1} \\ &= \mathbf{PD}^{20}\mathbf{P}^{-1} \end{aligned}$$

Note that $\mathbf{D}^{20}$ is easy to find.

Additional fact

If $\mathbf{A}$ can be written as $\mathbf{PDP}^{-1}$, i.e. $\mathbf{A} = \mathbf{PDP}^{-1}$

then, $\mathbf{A}^{-1} = [\mathbf{PDP}^{-1}]^{-1}$

$$= (\mathbf{P}^{-1})^{-1} \mathbf{D}^{-1} \mathbf{P}^{-1}$$

$$= \mathbf{PD}^{-1}\mathbf{P}^{-1}$$

$$\boxed{\mathbf{A}^{-1} = \mathbf{PD}^{-1}\mathbf{P}^{-1}}$$

$$\mathbf{D} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -5 \end{pmatrix}$$

$$\mathbf{D}^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & -1/5 \end{pmatrix}$$

Recall
 $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$

Example 19

A matrix is given as $\mathbf{B} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{pmatrix}$. It is known to have eigenvalues $\lambda = 1, 2, 3$ and corresponding eigenvectors

$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$. Find $\mathbf{B}^6$.

$$\left(\text{Ans: } \mathbf{B}^6 = \begin{pmatrix} 1 & 728 & 602 \\ 0 & 729 & 665 \\ 0 & 0 & 64 \end{pmatrix} \right)$$

Solution

$$\text{Let } \mathbf{D} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \mathbf{P} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$\mathbf{D}^6 = \begin{pmatrix} 1^6 & 0 & 0 \\ 0 & 2^6 & 0 \\ 0 & 0 & 3^6 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 64 & 0 \\ 0 & 0 & 729 \end{pmatrix}$$

Find $\mathbf{P}^{-1}$

$\vdots$

$$\mathbf{P}^{-1} = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\mathbf{B}^6 = \mathbf{P} \mathbf{D}^6 \mathbf{P}^{-1}$$

$\vdots$

Ans $\neq$

Example 20

Given that one of the eigenvectors of the matrix $A = \begin{pmatrix} 2 & -5 & 0 \\ 1 & a & 3 \\ 0 & 0 & 5 \end{pmatrix}$ is $\begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix}$, find matrices P and E such that $A^5 = PEP^{-1}$. (You are *not* required to find P^{-1} .)

$$\boxed{E = D^5}$$

$$\left(\text{Ans: } P = \begin{pmatrix} 1 & 5 & 15 \\ 1 & 1 & -9 \\ 0 & 0 & -32 \end{pmatrix}, E = \begin{pmatrix} -243 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3125 \end{pmatrix} \right)$$

Solution

Let λ_1 be the eval corr. to evec $\begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix}$

$$A \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix} = \lambda_1 \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix}$$

$\vdots$

$$\lambda_1 = 1, a = -4$$

$$A - \lambda I = \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix}$$

$$\det(A - \lambda I) = 0$$

$\vdots$

$$\lambda = 1, 5, -3$$

For $\lambda = 5$

$$A - \lambda I = \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix}$$

$$\begin{pmatrix} & & \\ & & \\ & & \end{pmatrix} \times \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix} = \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix}$$

For $\lambda = -3$

$\vdots$

$$\therefore P = \begin{pmatrix} 5 & \boxed{} & \boxed{} \\ 1 & \boxed{} & \boxed{} \\ 0 & \boxed{} & \boxed{} \end{pmatrix}, D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & -3 \end{pmatrix}$$

$$E = D^5 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 5^5 & 0 \\ 0 & 0 & (-3)^5 \end{pmatrix}$$

$\vdots$

$$= \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix}$$

***** END OF SECTION I *****